

A Systematic Mapping Study of Quantum Software Engineering Maturity to Guide a Brazilian Research Agenda

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Abstract

Context: Quantum Software Engineering (QSE) has emerged as a key field for developing reliable, maintainable, and scalable quantum applications. **Problem:** Despite its growth, QSE knowledge remains fragmented, lacking standardized methods, tools, and guidelines, and Brazilian participation in QSE research is still limited. **Objective:** This study characterizes the global state and maturity of QSE to inform a research agenda for the Brazilian community. **Method:** We conducted a systematic mapping study over four databases, selecting 275 primary studies published from 2002 to 2026 and classifying them by research strategy and predominant Software Engineering Body of Knowledge (SWEBOK) areas. **Results:** The corpus is mostly conference-based and concentrates on *Software Testing* (57), *Software Engineering Models and Methods* (39), and *Software Architecture* (33). Spain leads by country (48), Brazil appears with four studies, and *proposals of solution* (106) remain the largest strategy, although *evaluation and validation research* together account for 118 studies. **Conclusions:** These findings support a Brazilian QSE agenda organized around education and capacity building, empirical and interdisciplinary research, national venues and shared repositories, and dedicated funding and infrastructure for software-specific quantum research.

Keywords

Quantum Software Engineering, Research Agenda, Systematic Mapping Study

1 Introduction

The emerging field of Quantum Software Engineering (QSE) aims to systematize the development, testing, debugging, operation, and maintenance of quantum software systems [1, 13]. Prior secondary studies have mapped QSE topics, techniques, agendas, and regional initiatives [12, 20, 25]. As detailed in Section 2, our contribution is an updated primary-study map that combines research strategies with Software Engineering Body of Knowledge (SWEBOK) areas and uses this evidence to derive a Brazilian agenda. This fragmentation creates a practical problem because, without an up-to-date map of how QSE research has evolved, where it is produced, and which strategies and knowledge areas it covers, the Brazilian community has little evidence for defining priorities. In this context, a Systematic Mapping Study (SMS) is appropriate because it offers a broad and reproducible view of an emerging area and exposes underexplored topics that can ground an agenda.

This paper therefore presents a SMS of QSE and uses its findings to derive implications for a Brazilian research agenda. The map

covers 275 primary studies published from 2002 to 2026. It shows rapid growth after 2023, conference-driven dissemination, concentration in a few international hubs, limited Brazilian participation, a predominance of proposal of solution studies, and a focus on Software Testing, Software Engineering Models and Methods, and Software Architecture. The central contribution of this work is a Brazilian QSE agenda grounded in an international SMS, in which the mapping characterizes temporal evolution, geographic concentration, venues, research strategies, and SWEBOK areas, while the agenda translates these findings into directions for education, interdisciplinary research, national venues, and funding.

The remainder of this paper is organized as follows: Section 2 contrasts this study with prior secondary studies; Section 3 presents the protocol, selection process, and classification strategy; Section 4 reports and discusses the results; Section 5 derives the Brazilian research agenda from the mapping findings; and Section 6 concludes the paper.

2 Related Work

Prior QSE secondary studies have already mapped topics, challenges, and regional initiatives, so the novelty of this paper does not rest only on updating the publication window. De Stefano et al. [20] provide the closest global systematic mapping, Mandal et al. [12] review QSE techniques and sketch a general agenda, and Aparicio-Morales et al. [25] combine mapping and expert survey evidence for Latin America. Other influential works, such as Dwivedi et al. [5] and Piattini et al. [16], are better understood as narrative or pragmatic overviews because they do not report a full systematic protocol. We treat these works as related work rather than as data sources, since our corpus is restricted to primary studies.

Table 1 summarizes the comparison. Our study differs along four dimensions. First, it analyzes 275 primary studies from 2002 to 2026, which extends the coverage of previous systematic studies. Second, it uses a stricter lifecycle-oriented view of QSE, reducing the risk of conflating QSE with broader quantum-computing applications. Third, it combines research strategy and SWEBOK area to interpret empirical maturity rather than only topic frequency. Finally, it uses this international map as the empirical basis for a Brazilian research agenda, whereas previous agendas are either general to the field or regional without a structured connection to Brazilian Software Engineering (SE) priorities.

Conceptually, our contribution is the analytical bridge between global maturity indicators and national research priorities. Aparicio-Morales et al. [25] combine a Latin American mapping with an expert survey to characterize regional initiatives and barriers. In

contrast, our joint analysis of research strategies and SWEBOK areas identifies where topic coverage and empirical support remain uneven, while the geographic and venue results identify possible capacity and community entry points. The novelty therefore lies not in corpus size alone, but in making the transformation from international mapping findings to cautious Brazilian research implications explicit and traceable.

3 Research Method

In accordance with guidelines [10], this SMS was organized into three main phases: planning, conducting, and reporting. Each phase is described in this section.

3.1 Planning

The goal of this study is twofold: (i) to investigate the state of the art and evolution of QSE research and (ii) to define a research agenda for the Brazilian community.

To this end, the study is guided by one overarching question: *how can the current state and maturity of QSE research guide future research directions for the Brazilian community?* We operationalize through five subquestions, each oriented toward an analytical or maturity dimension that informs the agenda:

- **RQ1:** How has QSE research activity evolved over time, and what does this trajectory indicate about the field's maturity?
- **RQ2:** Which countries and institutions lead QSE research, and how is Brazil positioned within this landscape?
- **RQ3:** Through which venues is QSE knowledge disseminated, and what does this reveal about the consolidation of the QSE community?
- **RQ4:** What research strategies characterize QSE studies, and to what extent is the field supported by empirical evidence?
- **RQ5:** Which SWEBOK knowledge areas concentrate QSE research, which remain underexplored, and how do research strategies distribute across them?

The search string followed the PICO framework [10], in which quantum computing terms represented the population and SWEBOK-derived SE terms represented the intervention. The complete search string is available in the Artifact Availability section. Taken together, the answers to RQ1–RQ5 characterize the global state and maturity of QSE and provide the evidence base from which we derive the Brazilian research agenda (Section 5). In this study, we searched ACM Digital Library, IEEE Xplore, Scopus, and Web of Science because they cover peer-reviewed computer science and engineering literature.

We performed the search by applying the search string to the metadata (i.e., title, abstract, and keywords) of each retrieved publication. We also defined and applied inclusion and exclusion criteria to filter publications relevant to the study. A study was included only when it satisfied all of the following inclusion criteria: (i) it addresses QSE, that is, the engineering of quantum software systems rather than quantum algorithms or quantum information in isolation; (ii) it is a primary study; and (iii) it was published as a conference paper or journal article. A study was excluded when it met any of the following exclusion criteria: (i) it is not available online; (ii) it is not written in English; (iii) it is a secondary or tertiary study, such as a survey, review, or mapping; or (iv) it does not

address software development aspects of quantum computing (e.g., works focused only on quantum algorithms, quantum information, or hardware).

3.2 Conducting

Table 2 presents the number of studies after each filtering step of the SMS protocol. We conducted the search in two rounds. The first round, carried out between February and April 2025, retrieved 3294 publications and, after screening against the inclusion and exclusion criteria, yielded 167 primary studies. To incorporate the most recent literature, a second round extended the coverage to studies published up to 2026, retrieving 1263 additional records, from which 108 new primary studies were selected after removing duplicates and applying the same criteria. The combined corpus comprises 275 primary studies published between 2002 and 2026, as detailed per source and round in Table 2. We used Parsifal, Microsoft Excel, and Google Colab to support screening, data tabulation, exports, and chart generation.

From each selected primary study, we extracted title, authors, year, venue, country, institution, source database, research strategy, and predominant SWEBOK area. For research strategies, we followed the categories proposed by Wieringa et al. [23]: evaluation research, proposal of solution, validation research, philosophical paper, opinion paper, and personal experience paper. For SWEBOK, each study received one predominant area [22]. When studies were too general to fit into a specific area, we categorized them as *Software Engineering in General*. The protocol did not include a separate adjudication step for borderline SWEBOK cases. Their assignment therefore reflects the classifier's judgment under the predominant area rule, while *Software Engineering in General* was reserved for studies without a clear lifecycle focus. This limitation is considered in Section 4.6.

The research team comprised three researchers: two senior SE researchers who reviewed and validated the protocol and one researcher with the strongest Quantum Computing (QC) background who performed screening, extraction, and human classification. Large Language Models (LLMs) supported only operational standardization and inconsistency checks; no study data or classifications were generated by LLMs.

Because the corpus contains 275 studies and the workshop page limit prevents a full listing in the paper, the complete metadata, raw data, consolidated data, Jupyter Notebook, and the link to the interactive dashboard are provided in the replication package through the Artifact Availability section. The dashboard allows readers to filter the studies by different fields and provides a complementary way to explore the results, improving transparency, reproducibility, navigability, and access to the data beyond direct spreadsheet inspection.

4 Results and Discussion

This section presents and analyzes the results obtained for each research subquestion, providing an overview of the current state of QSE research. These findings provide the basis for the Brazilian research agenda discussed in Section 5.

Table 1: Secondary studies on QSE most related to ours, by methodology and coverage

Study	Type	Period	Databases	Studies	Scope / focus
De Stefano et al. [20]	Systematic mapping	2018–2023	4	113	Global; topics, venues, authors
Mandal et al. [12]	Systematic review	2004–2023	6 + manual	56	Global; QSE techniques and a generic agenda
Aparicio-Morales et al. [25]	Mapping + expert survey	2012–2023	2	91	Latin America; initiatives and gaps
de Paiva et al. [14]	Rapid review (in progress)	≤2024	1	7*	Brazil; SWEBOK-based technology gaps
Dwivedi et al. [5]	Narrative survey	n/a	n/a	n/a	Global; lifecycle and programming languages
Piattini et al. [16]	Pragmatic survey	n/a	n/a	n/a	Global; practical QSE challenges
This study	Systematic mapping	2002–2026	4	275	Global maturity (strategy × SWEBOK) and a Brazilian agenda

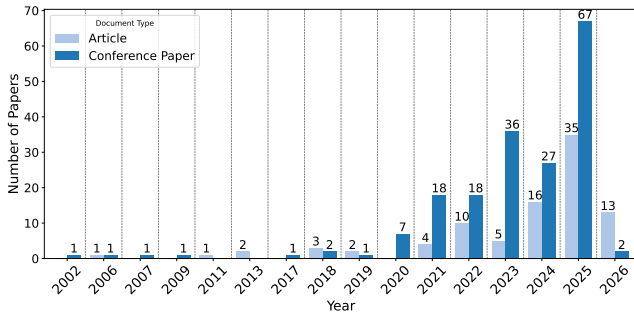
* Pilot content-analysis sample of an ongoing rapid review; n/a: not applicable to non-systematic overviews that do not define a search protocol.

Table 2: Primary studies selected per source and search round

Source	Round 1 selected	Round 2 selected	Total selected
ACM Digital Library	1	33	34
IEEE Xplore	0	45	45
Scopus	140	25	165
Web of Science	26	5	31
Total	167	108	275

4.1 Temporal Evolution and Maturity (RQ1)

Figure 1 shows three periods: low production from 2002 to 2020, initial growth in 2021–2022, and sharp acceleration from 2023 to 2025. The corpus grew from 41 studies in 2023 to 43 in 2024 and 102 in 2025; the 15 studies mapped for 2026 are a partial count and should not be read as a decline. Since 2020, conference papers have consistently outnumbered journal articles, suggesting that QSE is still using fast dissemination channels while its specialized venues consolidate. For Brazil, this creates an opportunity to enter the field while priorities, venues, and methods are still being shaped.

**Figure 1: Conference papers and journal articles per year**

4.2 Leading Countries, Institutions, and Brazil's Position (RQ2)

Figure 2 counts each country at most once per selected study and shows concentration in a small set of countries: Spain (48), the United States (38), Japan (37), Germany (34), China (32), and Finland (31). Table 3 summarizes the leading countries and institutions, indicating that QSE is structured around a limited set of international hubs rather than evenly distributed across regions. The most recurrent institutions include the University of Castilla-La Mancha (25), University of Jyväskylä (20), University of Extremadura (19), and Simula Research Laboratory (16). These counts indicate sustained research activity in a few groups, although the mapping does

not establish causal links between institutional structure, funding, and publication output.

Brazil appears with four selected studies and ranks 18th among the 48 countries represented in the corpus. These studies address non-functional requirements [19], test smells [21], testing in Qiskit algorithms [17], and future directions for testing and debugging [11]. This limited presence motivates agenda directions centered on SE expertise rather than competition on quantum hardware.

4.3 Publication Venues and Community Consolidation (RQ3)

Figure 3 shows that the leading venues are mostly conferences and workshops: the *International Conference on Quantum Computing and Engineering* (30), *International Workshop on Quantum Software Engineering* (14), *International Conference on Automated Software Engineering* (13), *International Conference on Quantum Software* (13), and *International Workshop on Quantum Software Engineering and Technology* (8). Conference papers dominate the corpus (183) over journal articles (92), but established SE journals also appear, including *Transactions on Software Engineering and Methodology* (10), *Information and Software Technology* (7), and *IEEE Transactions on Software Engineering* (6). The spread across more than one hundred venues suggests an emerging, not fully settled, community and supports national entry points such as workshops, tutorials, and special tracks.

4.4 Research Strategies and Empirical Maturity (RQ4)

This RQ analyzes research strategies using the six Wieringa et al. categories described in Section 3. Figure 4 combines research strategies and SWEBOK areas. *Proposal of solution* is the largest strategy (106), followed by *evaluation research* (65), *validation research* (53), *opinion papers* (33), *philosophical papers* (11), and *personal experience papers* (7). The 118 evaluation and validation studies suggest emerging empirical maturation, but the dominance of proposals over each individual empirical strategy still indicates a field more solution-driven than evidence-driven.

4.5 SWEBOK Areas and Research Gaps (RQ5)

To answer RQ5, each primary study was assigned to one predominant SWEBOK area [22]; broad studies were grouped as *Software Engineering in General*. This exclusive scheme enables comparison but may underrepresent multi-area studies. The most frequent areas are *Software Testing* (57), *Software Engineering Models and Methods* (39), *Software Architecture* (33), and *Software Construction* (32). This concentration reflects attention to foundational abstractions,

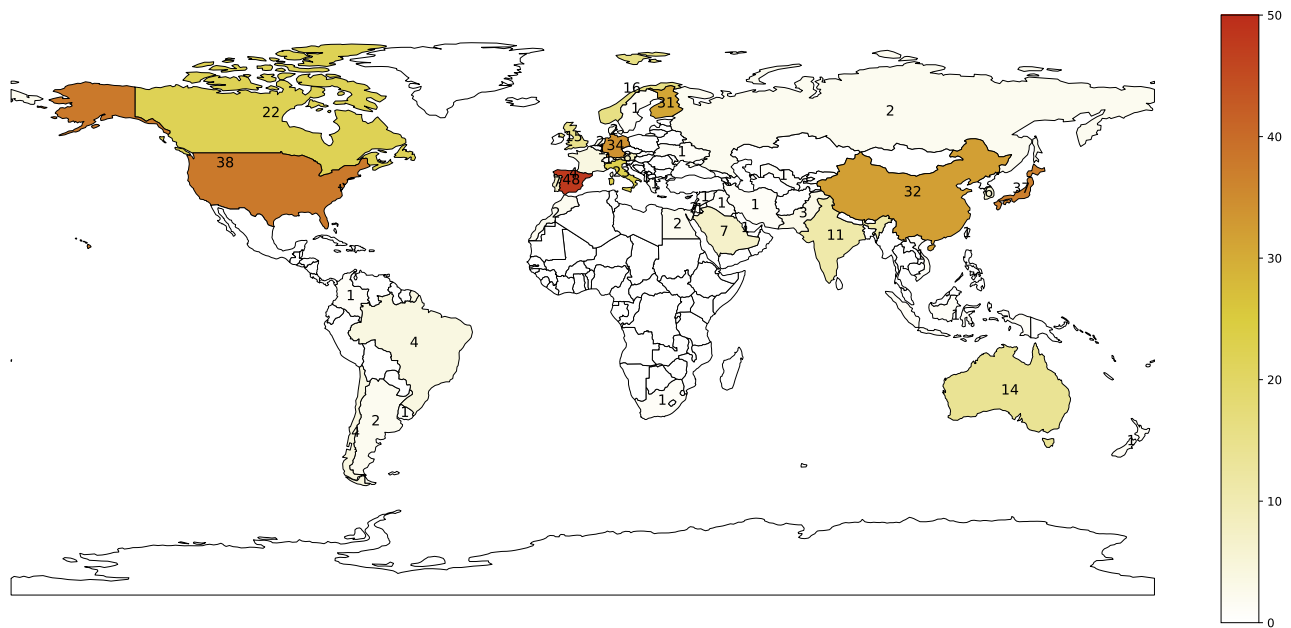


Figure 2: Global distribution of publications by country

Table 3: Leading countries and institutions in the selected corpus

Country	Studies	Most recurrent institutions by study-institution occurrences
Spain	48	University of Castilla-La Mancha (25); University of Extremadura (19); aQuantum (6)
United States	38	University of California (3); University of Maryland (3); City University of New York (2); College of William & Mary (2); Duke University (2)
Japan	37	Kyushu University (18); National Institute of Informatics (14); NTT Computer and Data Science Laboratories (3)
Germany	34	University of Stuttgart (6); Technical University of Munich (4); Lancaster University Leipzig (3)
China	32	Tsinghua University (7); Beihang University (5); Institute of Software, Chinese Academy of Sciences (5)
Finland	31	University of Jyväskylä (20); University of Oulu (13); Lappeenranta-Lahti University of Technology (5)
Brazil	4	University of Sao Paulo (2); Federal University of Alagoas (1); Federal University of Catalao (1); Federal University of Ceara (1); Pontifical Catholic University of Rio de Janeiro (1)

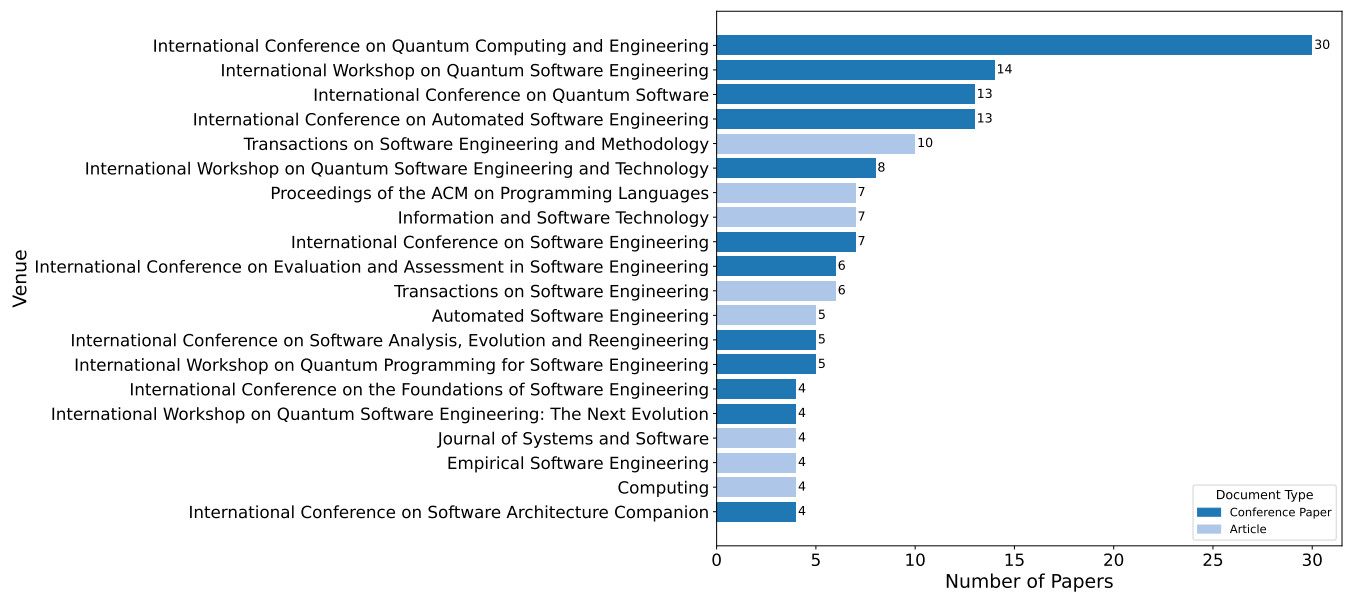


Figure 3: Top 20 venues by number of studies, grouped by document type



Figure 4: Relationship between research strategies and SWEBOK areas among primary studies

architectural models, construction techniques, and testing strategies. Less frequent or absent areas, including maintenance, process, management, requirements, economics, professional practice, and configuration management, remain opportunities for distinctive QSE contributions.

The cross-analysis shows that *Software Engineering Models and Methods* has the most proposal-of-solution studies (25), whereas *Software Testing* is the most empirically balanced area, with 18 evaluation [8] and 16 validation [21] studies, including work on mutation testing [8] and test smells [21]. For the Brazilian agenda, underexplored lifecycle and management areas are therefore topic gaps and opportunities to produce stronger empirical evidence.

4.6 Threats to Validity

We discuss validity using construct, internal, external, and conclusion validity for a secondary study [24]. *Construct validity*: the database search may miss studies using non-standard vocabulary, full-text-only QSE framing, or non-indexed venues; the lifecycle-based inclusion criteria reduce but do not eliminate judgment in separating QSE from broader quantum-computing work. *Internal validity*: screening, extraction, and classification involved interpretation, were not independently double-coded, and did not use an

agreement statistic; LLMs supported only standardization and inconsistency checks, while classifications remained human decisions. This concentration in one classifier remains a limitation. Published taxonomies, the predominant area rule, consistency checks, and classifications reported for each study improve transparency and auditability, but do not substitute for independent coding. *External validity*: the 2002–2026 window is affected by indexing lag, especially for 2026, and indexed English-language venues may underrepresent Brazilian work. *Conclusion validity*: the study reports frequencies rather than evidence quality or causal effects, so maturity and agenda claims are descriptive interpretations tied to mapping findings.

5 Implications for a Brazilian QSE Research Agenda

The answers to RQ1 through RQ5 describe the structure of the mapped field. We derive the Brazilian agenda from this evidence rather than from a predefined policy narrative. The corpus shows rapid growth, geographic concentration, dissemination primarily through conferences, a strong presence of solution proposals, and uneven coverage of SWEBOK areas. Brazil appears in only four selected studies. The agenda therefore presents research implications for entering a field whose priorities, venues, and empirical

Table 4: Compact traceability between mapping findings and agenda directions

Finding	RQ	Implication for Brazil	Agenda direction
Rapid growth and partial 2026 count	RQ1	Brazil can still engage while QSE priorities and training paths are forming.	Education and capacity building
Concentrated production and four Brazilian studies	RQ2	Isolated publications are unlikely to create critical mass.	Research groups and collaboration
Conference-driven and dispersed venues	RQ3	National entry points can connect QSE with established SE forums.	Workshops, tracks, and shared artifacts
Proposal-driven strategies and uneven SWEBOK coverage	RQ4–RQ5	Brazil can contribute empirical evidence in underexplored lifecycle and management areas.	Empirical research, funding, and infrastructure

standards are still being shaped. Table 4 summarizes this derivation by linking each agenda direction to the corresponding mapping findings.

Education and capacity building. The temporal pattern in RQ1 and Brazil’s limited participation in RQ2 suggest that education is a precondition for a sustainable Brazilian QSE agenda. Training should connect mathematical and quantum foundations with SE life-cycle concerns through courses, open materials, and hybrid-system projects. Because regional evidence identifies subject complexity and a lack of experts as participation barriers, Portuguese-language resources, instructor development, and tools such as Ket could lower the entry barrier [18, 25].

Empirical and interdisciplinary research. RQ4 shows that proposal of solution is the largest single strategy, despite substantial evaluation and validation activity. Brazilian groups could prioritize experiments, replications, benchmark-based evaluations, and comparative tool studies that combine SE, quantum-computing, and empirical-method expertise. The four Brazilian studies do not reveal the number of groups, grants, or supervisors, but regional evidence identifies limited funding and collaboration as barriers [25]; shared protocols could help small teams pool participants, hardware time, and expertise.

Research groups, collaboration, and community. The concentration in RQ2 suggests that stable groups may give continuity to QSE research, without establishing a causal link between institutional structure and publication output. Brazil could connect quantum-computing, SE, and empirical-methods groups through explicit QSE projects, co-supervision, and reusable artifacts such as benchmarks and execution metadata; the lifecycle-oriented aQuantum/UCLM profile illustrates this research focus [2]. The conference-driven dissemination in RQ3 supports lower-cost entry points within established events rather than an immediate standalone venue. Tracks, tutorials, and artifact demonstrations could connect Brazilian work to international forums, while the *I Brazilian Workshop on Quantum Software Engineering*, co-located with the *Brazilian Conference on Software* in 2026, provides an initial national example anchored in SE.

Funding, infrastructure, and software tooling. The concentration in RQ2 and the gaps in RQ5 support funding that explicitly targets software-engineering problems in quantum software. Brazilian initiatives indicate broader quantum-technology activity, not consolidated QSE funding [6, 7]; priorities could include testing infrastructure, reproducible benchmarks, cloud execution, and industry-academia tooling. Hardware access is a quantum-specific constraint: free cloud quotas are limited, while sustained execution involves credentials, queues, noise, and possible costs even when simulators support preliminary experiments [9, 15, 25]. Funding

could support protocols that compare ideal and noisy simulation with budgeted hardware runs. Broader national infrastructure such as CIQUANTA-PB and the CBPF laboratory may enable this work, but should not be presented as established QSE programs [3, 4]. These directions are evidence-linked research implications, not a claim that Brazil can reproduce leading countries’ trajectories or a prescriptive policy program.

6 Conclusion

This study characterized the global state and maturity of QSE through an SMS of 275 primary studies published from 2002 to 2026 and used the evidence to derive implications for the Brazilian SE community. The results show rapid growth, concentration in a limited set of countries and institutions, and incipient Brazilian participation, with four selected studies. The mapping also indicates an area still consolidating: conference papers dominate, proposal of solution remains the largest single strategy, and the most frequent SWEBOK areas are *Software Testing*, *Software Engineering Models and Methods*, *Software Architecture*, and *Software Construction*. Evaluation and validation research already form an important part of the corpus, but underexplored lifecycle and management areas remain opportunities for future work. By answering RQ1–RQ5, the study links the mapped state and maturity of QSE to Brazilian research directions: education and capacity building, empirical and interdisciplinary research, national venues and shared artifacts, and software-specific funding, infrastructure, and tooling. The agenda should be read as research implications rather than a prescriptive policy program. Its contribution is to make Brazilian directions traceable to the international state of the art while acknowledging that the mapping reports publication patterns and classification frequencies, not causal explanations or evidence quality.

Artifact Availability

The **replication package** provides selected-study metadata, raw and consolidated data, and figure-generation notebooks; the **interactive dashboard** supports complementary exploration.

Acknowledgments

This research was partially supported by the Federal Institute of Southeast Minas Gerais (IF Sudeste MG) and Brazilian funding agencies: CAPES, CNPq (Grants 406089/2025-6), and FAPEMIG (Grant APQ-01488-24). In addition, the authors used ChatGPT for grammar revision, English translation, and screening-phase data checks, and ChatGPT and Claude Code to support the analysis notebook and interactive dashboard. The authors critically reviewed, validated, and refined all AI-assisted outputs.

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